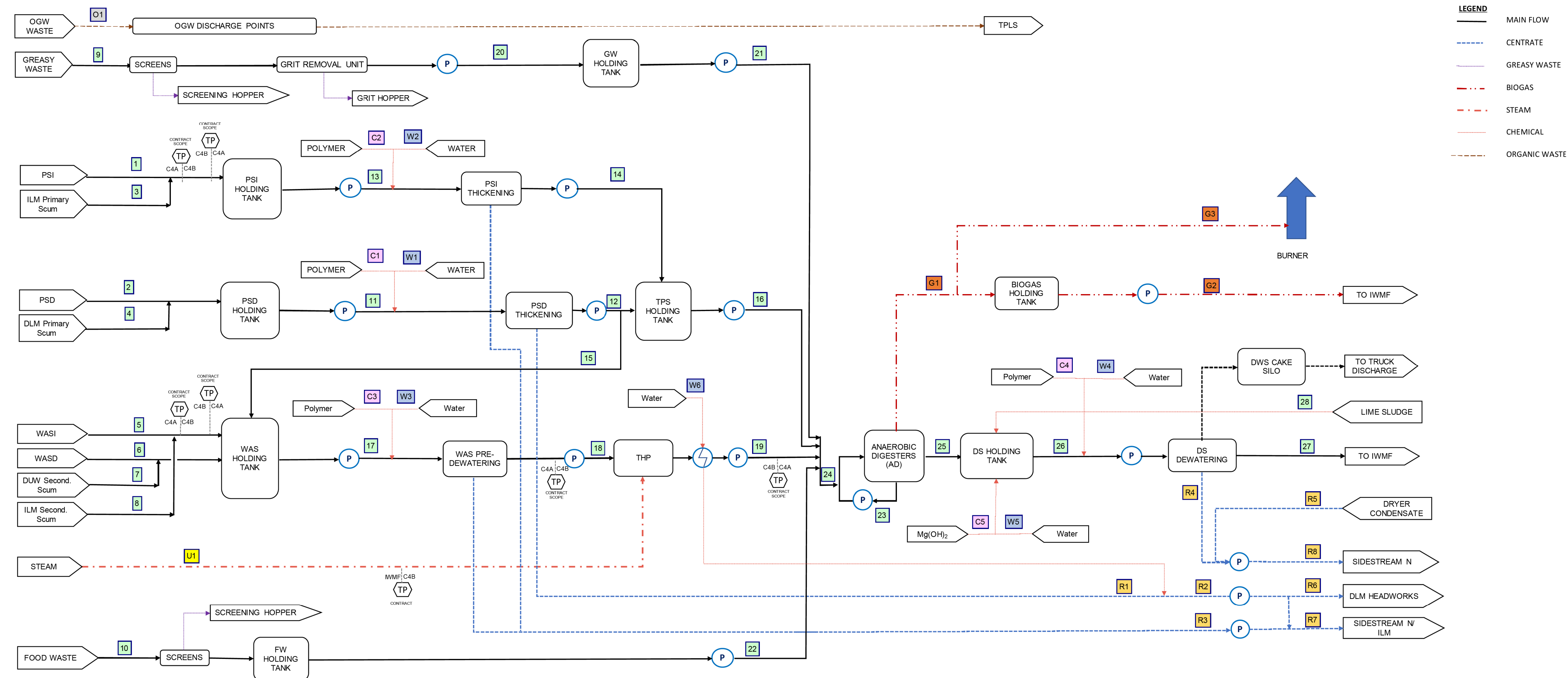




STREAM		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28
		PSI	PSD	ILM Primary scum	DLM Primary scum	WASI	WASD	DLM Secondary Scum	ILM Secondary Scum	Greasy Waste	Food Waste	PSD Thickener Inlet	PSD Thickener Outlet	PSI Thickener Inlet	PSI Thickener Outlet	PSD Transfer to THP	TPS Holding Tk Out.	WAS Pre-dewatering Inlet	WAS Pre-dewatering Outlet	THP Outlet	GW to Holding Tank	GW Feed to Digester	Food Waste to Digester	AD Mixing/ Circulation	AD Inlet	DS to DS HT	DS Dewatering Inlet	Dewatered Sludge to IWMF	Lime Sludge
Component Description (for D/S)		Tank	Tank			Tank						Pump	Centrifuge	Pump	Centrifuge		Pump	Pump	Centrifuge	Pump	Pump	Pump	Pump	Pump		Tank	Pump	Centrifuge	
Number of Duty Units	#	2	2	--	--	2		--	--	2	2	9	9	2	2		1	9	9	1	2	1	1	20	--	2	4	4	--
Number of Standby Units	#	0	0	--	--	0		--	--	0	0	5	5	2	2		2	2	2	2	2	2	2	20	--	0	2	2	--
Minimum Daily TSS Mass Load	tDS/d																												
Annual Average TS Mass Load	tDS/d	50	166	0.045	0.045	48	73	0.045	0.045	10.2	81	166	153	50	47	45.8	154.2	168	150.8	150.8	10.2	10.2	81.0		396.1	220.1	251.3	238.8	0.7
Maximum Monthly TS Mass Load	tDS/d	65	214	0.054	0.054	77	85	0.054	0.054	10.2	81	214	197	65	62	59.1	199.7	221	198.7	198.7	10.2	10.2	81.0		489.6	276.2	313.0	297.4	0.9
Maximum Day TS Mass Load	tDS/d	130	339			94	121																						
Minimum Daily TSS concentration	%DS																												
Average TS concentration	%DS	1.7	0.85	0.045	0.045	0.75	0.67	0.045	0.045	3.4	13.5	0.85	8.0	1.6	8.0	8.0	8.0	0.92	16.0	13.6	3.4	3.4	13.5		10.1	5.6	6.4	28.0	1.6
Maximum TS concentration	%DS	2.0	1.1	0.045	0.045	1.2	0.78	0.045	0.045	3.4	13.5	1.1	8.5	2.0	8.5	8.0	8.5	1.3	18.0	13.6	3.4	3.4	13.5		10.1	5.7	6.4	33.0	1.8
Minimum TS concentration	%DS	1.2	0.70			0.54	0.53																						
Minimum Daily Flow	m³/d																												
Annual Average Flow @ Avg. conc.	m³/d	3,000	19,500	100	100	6,450	10,900	100	100	300	600	19,600	1,906	3,100	593	572	1,927	18,122	942	1,108	300	300	600		3,935	3,935	3,947	1,066	42
Maximum Monthly Flow @ Avg. conc.	m³/d	3,913	25,161	120	120	10,294	12,563	120	120	300	600	25,301	2,462	4,033	773	738	2,496	23,887	1,242	1,461	300	300	600		4,857	4,857	4,870	1,328	51
Hydraulic Peak Flow (instantaneous)	L/s	90	462			146	207			125	14	293	37	47	11	9	29	276	13	17	25	7	7		60			16	4.7

STREAM		U1	R1	R2	R3	R4	R5	R6	R7	R8	G1	G2	G3	C1	C2	C3	C4	C5	W1	W2	W3	W4	W5	W6	O1
		Steam	TCD	TCD & HWR	TCI & DCW	DC	Dryer Centrate (SC) from IWMF	TCD/HWR to DLM¹	TCD/DCW to ILM or Sidestream N	DC/SC to Sidestream N	Biogas generated	Biogas generated to IWMF	Biogas generated to Flare	Polymer: PSD Thickening	Polymer: PSI Thickening	Polymer: WAS Pre-dewatering	Polymer: DS Dewatering	Mg(OH)₂	PSD poly. Prep	PSI poly. Prep	WAS poly. Prep	DS poly. Prep	Mg(OH)₂ Prep	HS and Proc Gas Cooling	OGW To TPLS
Component Description (for D/S)				Pumps	Pumps					Pump		Blower	Burner					Pumps							
Number of Duty Units	#	--	--	3	2	--	--	--	--	2	--	2	6	--	--	--	--	1	--	--	--	--	--	--	--
Number of Standby Units	#	--	--	3	2	--	--	--	--	2	--	2	2	--	--	--	--	2	--	--	--	--	--	--	--
Annual Average TS Mass Load	tDS/d	0.0	13.3	13.3	19.2	12.6	0.0	4.1	32.5	12.6	--	--	--	--	--	--	--	8.0	--	--	--	--	--	--	--
Maximum Monthly TS Mass Load	tDS/d	0.0	17.1	17.1	25.3	15.7	0.0	5.4	42.5	15.7	--	--	--	--	--	--	--	9.4	--	--	--	--	--	--	--
Average TS concentration	%DS	0.0	0.07	0.05	0.09	0.26	0.00	0.05	0.08	0.24	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
Maximum TS concentration	%DS	0.0	0.07	0.05	0.09	0.26	0.00	0.05	0.08	0.24	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--
								Nm³/d	Nm³/d	Nm³/d															
Annual Average Flow @ Avg. conc.	m³/d		18,375	26,720	21,235	4,895	428	8,345	39,610	5,323	237,532	237,532	237,532	0.88	0.27	2.68	4.02	11.49	680	205	1,340	2,011	8.00	8,345	4,968
Maximum Monthly Flow @ Avg. conc.	m³/d		23,719	34,719	27,942	6,052	538	11,000	51,661	6,590	287,725	287,725	287,725	1.14	0.35	3.53	5.01	13.51	878	267	1,766	2,504	9.40	11,000	9,936
Hydraulic Peak Flow (instantaneous)	L/s			402	323		6	127		80								89							
Annual Average Mass Flow @ Avg. conc.²	t/d	165.8	--	--	--	--	--	--	--	--	--	--	--	0.66	0.20	2.01	3.02	7.96	--	--	--	--	--	--	--
Maximum Monthly Mass Flow @ Avg. conc.²	t/d	218.6	--	--	--	--	--	--	--	--	--	--	--	0.86	0.26	2.65	3.76	9.37	--	--	--	--	--	--	--
Peak Mass Flow (instantaneous)	kg/s	3.7	--	--	--	--	--	--	--	--	5.07	5.07	5.07	--	--	--	--	--	--	--	--	--	--	--	--

Notes:
1. The default will be to pump TCD to ILM and HWR to DLM, if required to maximise recovery the TCD can be pumped to DLM
2. Steam - tonnes of steam per day; Polymer & Magnesium hydroxide - tonnes of powder per day.

TENDER

T1	20/05/22	ISSUED FOR TENDER C4B	CN	GOT
T3	11/12/20	ISSUED FOR VARIATION	CN	GOT
T2	21/10/19	ISSUED FOR ADDENDUM 3	CN	GOT
T1	16/08/19	ISSUED FOR TENDER - C4A	CN	GOT
REV	DATE	DESCRIPTION	REVIEWED	AUTHORISED

CLIENT:



PROGRAMME MANAGER:



CONSULTANT:



PROJECT TITLE:

TUAS WATER RECLAMATION PLANT (TWRP)
DTSS PHASE 2

DRAWING TITLE

PFD
BIOSOLIDS
PHASE-1

DRAFTED: AM	DESIGNED: KV	SHEET SIZE: A1
DRAFTING CHECK: ASR	DESIGN CHECK: CN	SCALE: NTS
DATE: 16/08/19	REVIEWED: CN	PBS: 131101
CONTRACT:	DRAWING NUMBER: PUB-DTSS2-TWRP	
	SHEET NUMBER: WSB01-P-7001	REVISION: T1
THIS DRAWING IS COPYRIGHT		